

OMAR LEMKECHER

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EDUCATION

- Bachelor of Science in **Aerospace Engineering** at the **University of California, Los Angeles** Expected Graduation: **2028**

SKILLS

- **Propulsion & Fluids** : N_2O thermodynamics, discharge coefficient modeling, Pressure Vessel design, CFD Basics
- **Structures & Mechanical Analysis** : FEA (ANSYS, SolidWorks), Stresses Analysis (Brackets, Endcaps, Bolt Analysis)
- **Software & Tools** : Python, C++, SolidWorks, Fusion360, Onshape, ANSYS
- **Machining & Prototyping** : CNC machining, CAM, 3D printing, hydrostatic testing, instrumentation integration
- **Interests** : Big Science Fiction reader, Soccer & Volleyball aficionado, Sweet potato fries devourer

EXPERIENCES

Hybrid Propulsion Feed Systems Lead - Rocket Project at UCLA

2025-Present

- Lead engineer for the design, analysis and test of Valkyrie's **Feed System**, a **600 lbf N_2O /HTPB SRAD Hybrid Rocket** that successfully launched to **13,300 ft** and fully **successfully recovered**
- Designed full oxidizer feed system (**750 psi MEOP**), including plumbing, fittings interfaces, and pressure relief strategy
- Performed discharge coefficient calculations and flow modeling to characterize pressure losses and transient behavior
- Designed and **FEA validated** lightweight oxidizer tank endcaps and brackets under internal pressure and bolt preload
- Conducted analysis on bolt bearing stresses and bracket bending to meet required margin of safeties
- Optimized tank ullage volume through **test-data-driven iteration** to maximize delivered impulse and combustion stability
- Manufactured all components using both manual and CNC machining across a rang

ADCS Lead - BruinSpace

2025- Present

- Leading end-to-end design and development of the Attitude Determination and Control System for a **3U+ CubeSat**
- Designed and built dedicated test setups for **reaction wheel synchronization** and rotation validation, as well as **magnetorquer** calibration and performance testing

PROJECTS

High-Pressure Vessel Design for Hybrid Rocket Oxidizer Feed System

2025-2026

- Designed and validated an aluminum **pressure vessel** assembly, 50" cylindrical tank and custom endcaps for a N_2O /HTPB hybrid rocket oxidizer feed system operating at 750 psi MEOP.
- Performed analytical **thin wall hoop and longitudinal stress calculations** by hand, then validated by **FEA shell** simulations on **Ansys** to confirm structural margins under full operating pressure applying an industrial margin of safety.
- Conducted a multi-objective design study optimizing simultaneously for pressure load resistance and mass reduction on endcaps and its bolts, applying **Von Mises yield criterion** as the failure condition across iterative FEA runs.
- Full structural **validation** was completed through **hydrostatic pressure testing** of the physical assembly, confirming correlation between analytical predictions, simulation outputs, and real-world load behavior.

66th Parabolic Flight Campaign of the French National Space Studies Center (CNES)

2023

- Designed and executed a series of **microgravity experiments** covering thermodynamics, centrifugal forces, and Newton's Laws validation. Following a one-year preparation and selection process, two team members flew to **CNES headquarters** in Bordeaux (France) to conduct the experiments aboard a **Zero-G flight**.

Young Searchers Prize, Project X

2023 & 2024 & 2025

- Annual engineering competition centered around a specific theme. Led two award-winning projects, a Space Probe concept (2023) and a Zero-G experimental system (2024). Selected in 2025 to present both projects at the national closing ceremony.

National Informatics Olympiads

2023

- Developed algorithmic solutions to qualify for the International Olympiad in Informatics (IOI). Ranked 18th nationally in Tunisia in the selection process.